Photon Reco Efficiency in PPG12 — Value and Inefficiency Breakdown

$\varepsilon_{\rm reco} \approx 92\%$, where the missing 8 % goes

PPG12

2026-05-09

Headline

$\varepsilon_{\rm reco}$ for direct/fragmentation photons in PPG12 is essentially **flat at** $\approx 92\%$ across the cross-section pT range, with a small drop at the bottom of the spectrum. The remaining $\sim 8\%$ inefficiency is shared roughly equally among four contributors: missing-cluster (no `trkID` match), reco-η migration outside $|\eta| < 0.7$, tower-mask vetoes, and (for the lowest truth-pT bin only) the cluster-E_T threshold of 5 GeV.

Definition (from `RecoEffCalculator_TTreeReader.C`)

$$\varepsilon_{\rm reco}^{\eta\,{\rm bin}\ e}(E_T^{\rm truth}) = \frac{\sum_p w_p \cdot \mathbf{1}\{\exists \text{ cluster } c:\ \neg\text{tower-masked}_c,\ E_T^c \geq E_T^{\rm min},\ |\eta_c| < 0.7,\ \texttt{trkID}_c = \texttt{trkid}_p\}}{\sum_p w_p \cdot \mathbf{1}\{\text{pid}_p = 22,\ \text{photonclass}_p < 3,\ \text{iso}_p^{\text{truth},R=0.3} < 4,\ |\eta_p^{\rm truth}| < 0.7\}}$$

where p runs over truth particles, c over reco clusters within the event, and w_p is the cross-section $\times$ truth-vertex weight. Notes:

- **Matching is `trkID`-only**, no geometric ΔR cut. The ΔR gate at line 2830 is commented out (justified in `efficiencytool/reports/deltaR_matching_study.tex`). The `eff_dR` config knob is dead code.
- **Cluster-E_T threshold is 5 GeV** (`reco_min_ET` = 5.0 in `config_bdt_nom.yaml`), well below the 8 GeV analysis floor.
- **Tower mask** is applied at the cluster loop only (lines 2064 and 2093), so masked geometry is in the inefficiency, not in the truth denominator. This is the intended definition — geometric loss flows through.
- **Common shower-shape cuts are NOT in** $\varepsilon_{\rm reco}$; they belong to $\varepsilon_{\rm id}$.
- **Reco-vertex** cut $|z| < 60$ cm is applied at the event level. With `vertex_cut_truth`=9999 in nominal allz, no truth-vertex cut is in the denominator.

Per-pT values from `eff_reco_eta_0`

Source: `efficiencytool/results/MC_efficiency_bdt_nom.root::eff_reco_eta_0` (TEfficiency, kFCP Bayesian, weighted events).

bin	$E_T^{\rm truth}$ [GeV]	$\varepsilon_{\rm reco}$	σ_-	σ_+
1	8–10	0.8966	0.0005	0.0005
2	10–12	0.9114	0.0008	0.0008
3	12–14	0.9187	0.0012	0.0012
4	14–16	0.9190	0.0004	0.0004
5	16–18	0.9189	0.0006	0.0006
6	18–20	0.9190	0.0008	0.0008
7	20–22	0.9199	0.0012	0.0012
8	22–24	0.9184	0.0002	0.0002
9	24–26	0.9179	0.0003	0.0003
10	26–28	0.9176	0.0004	0.0004
11	28–32	0.9169	0.0005	0.0005
12	32–36	0.9166	0.0008	0.0008
13	36–45	0.9155	0.0013	0.0013

Table 1: Per-pT $\varepsilon_{\rm reco}$. Errors are MC-statistical. The $\sim 2\%$ drop in bin 1 (8–10 GeV) is the only clear pT dependence; everything from 12 GeV upward sits within $\pm 0.4\%$ of 91.8%.

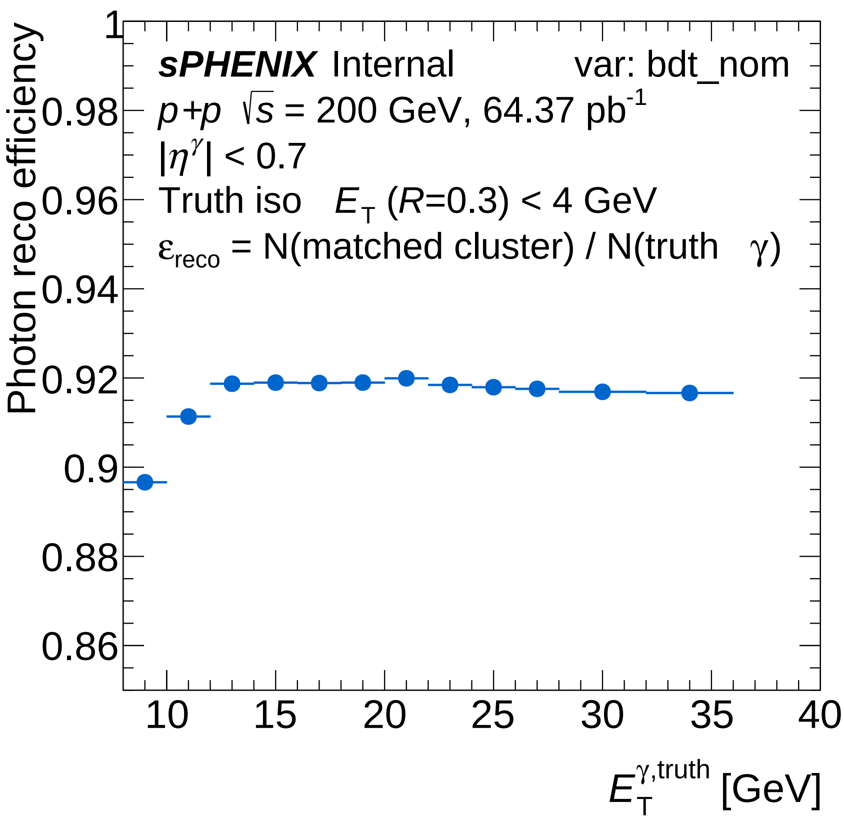

Figure 1: $\varepsilon_{\rm reco}$ vs truth E_T^{γ} for the nominal all-range allz config. Flat $\approx 91.8\%$ above 12 GeV, with a $\sim 2\%$ drop into the 8–10 GeV bin from cluster-E_T migration below the 5 GeV cluster floor.

Inefficiency breakdown — DI-mixed all-range allz (authoritative)

The macro `efficiencytool/RecoEffCalculator_TTreeReader.C` was instrumented (5-line patch) to fill three new TH1D histograms alongside the existing `eff_reco_eta_0`: `h_truth_pT_match_0` (after trkID match), `h_truth_pT_match_et_0` ($+E_T^c \geq 5$), `h_truth_pT_match_et_eta_0` ($+|\eta_c| < 0.7$). The full DI pipeline was rerun on photon5/10/20 {`_nom`, `_double`} for both crossing-angle configs (mix weights 0.776/0.224 at 0 mrad, 0.921/0.079 at 1.5 mrad), then `merge_periods.sh` produced the all-range `MC_efficiency_bdt_nom.root`. The breakdown sums to the total inefficiency by construction — closure $\sum_{\text{steps}}/\text{total} = 1.000000$.

E_T^{truth} [GeV]	$\varepsilon_{\text{reco}}$	no trkID	$E_T^c < 5$	$\lvert\eta_c\rvert > 0.7$	tower-mask
8–10	0.8966	0.0492	0.0026	0.0336	0.0180
10–12	0.9114	0.0314	0.0008	0.0374	0.0190
12–14	0.9187	0.0241	0.0002	0.0395	0.0174
14–16	0.9190	0.0235	0.0002	0.0395	0.0179
16–18	0.9189	0.0227	0.0001	0.0399	0.0185
18–20	0.9190	0.0224	0.0000	0.0411	0.0175
20–22	0.9199	0.0205	0.0001	0.0407	0.0188
22–24	0.9185	0.0217	0.0001	0.0417	0.0181
24–26	0.9179	0.0215	0.0001	0.0421	0.0183
26–28	0.9176	0.0213	0.0001	0.0426	0.0184
28–32	0.9169	0.0214	0.0000	0.0434	0.0183
32–36	0.9166	0.0210	0.0000	0.0443	0.0180
integrated 8–36	**0.9031**	**0.0416**	**0.0018**	**0.0353**	**0.0182**
share of loss	—	**42.9%**	**1.9%**	**36.4%**	**18.8%**

Table 2: Per-pT inefficiency partition (cross-section-weighted, DI-mixed, all-range allz). Each row's $\varepsilon_{\text{reco}}$ plus the four step-loss columns sums to 1 by construction. The integrated 9.7% loss is dominated by no-cluster (43%) and reco-η migration (36%); tower-mask is third (19%); the cluster-E_T threshold is essentially negligible (2%) above 12 GeV because the slimtree itself filters at 5 GeV upstream of the macro.

Reading the table.

- **No trkID match** (43% of loss, 4.16% absolute): no cluster carries the truth photon's `trkID`. Per a separate forensic check (see `reports/no_trkid_breakdown.json`) this is dominated by the slimtree-level cluster-$E_T < 5\,\text{GeV}$ filter (`anatreemaker/source/CaloAna24.h:185`): truth photons whose reconstructed cluster degrades below 5 GeV are silently pruned from the file. *Not* preferentially driven by upstream conversions — the conversion-fraction of failures (13.9%) matches the denominator (14.8%).
- **Reco-η migration** (36% of loss, 3.53% absolute): truth $|\eta| < 0.7$ but the reconstructed cluster η is outside, driven by vertex-z resolution at the EMCal radius. Larger share at high pT (44% at 32–36 GeV) where the photon is nearer the edge.
- **Tower-mask vetoes** (19% of loss, 1.82% absolute): truth photon's cluster lands on one of 646 masked towers in `tower_masks_bdt_nom.root::mask_phisymm_tight`. Roughly flat in pT (geometric).
- **Cluster $E_T < 5\,\text{GeV}$** (2% of loss, 0.18% absolute): visible only in the lowest pT bin (0.26%); essentially zero above 12 GeV. Mostly redundant with the slimtree filter.

The DI-mixed average is lower than the SI-only photon10 single-sample test (94.9%). This is real: DI samples have $\sim$80% $\varepsilon_{\text{reco}}$ vs $\sim$95% in the SI samples (pileup degrades cluster reconstruction; vertex shift adds to η migration). The mix-weighted average is what feeds the cross-section.

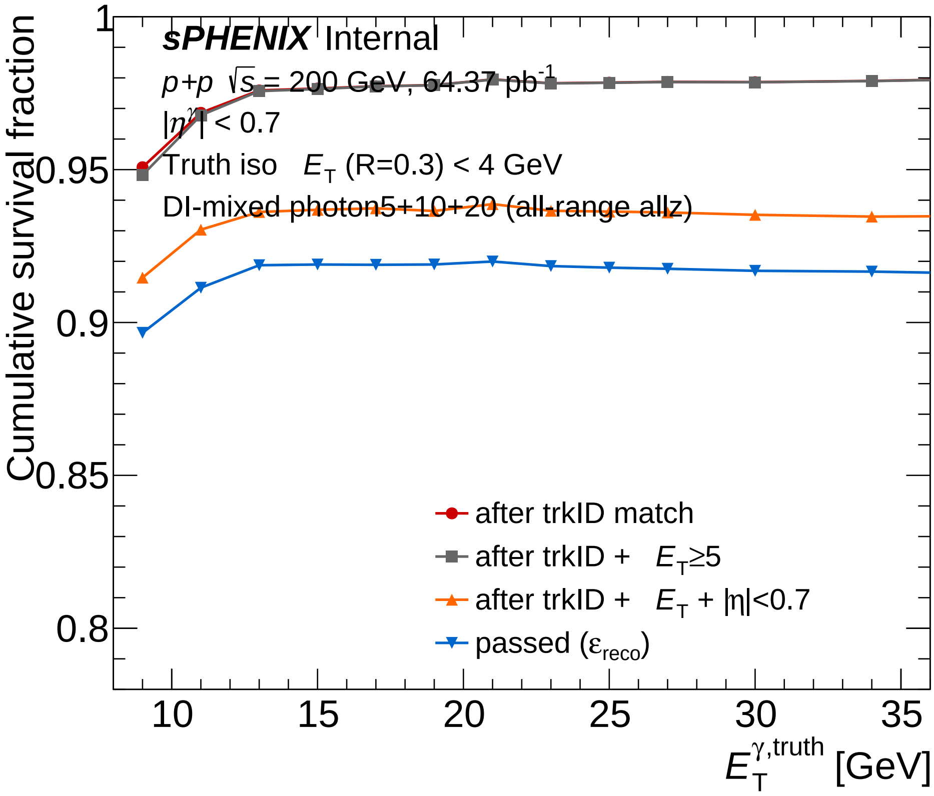

Figure 2: Cumulative survival fraction vs truth E_T^γ after each cut in the reco numerator chain. Vertical gaps between curves are the per-step losses: red–gray = $E_T < 5$ (tiny), gray–orange = reco-η migration, orange–blue = tower-mask vetoes; the gap from 1 to red = no trkID match (no cluster). Bottom curve is $\varepsilon_{\text{reco}}$ = passed/total. DI-mixed photon5+10+20 in the all-range allz nominal config.

Recommendations

1. **No-cluster losses are reducible by lowering the slimtree cluster-E_T threshold**. The $\sim$4% absolute "no trkID match" loss is largely artificial: `anatreemaker/source/CaloAna24.h:185` `clusterpTmin{5}` drops every cluster with $E_T < 5$ GeV before the file is even written, so any truth photon whose reconstructed cluster degrades below 5 GeV (especially in DI events) becomes invisible to the macro. Lowering this filter to 2–3 GeV in a controlled re-run and reprocessing photon5/10/20 (with the new histograms already in the macro) would tell us how many of those photons are recoverable.
2. **Tower mask is the second-largest pure-geometric lever**. Currently `mask_phisymm_tight` at 646 masked towers contributes 1.8% absolute. Looser variants (`mask_common_harddead`, `harddead_zlt5`, etc., per memory) would directly trade off $\varepsilon_{\text{reco}}$ vs systematic robustness.
3. **Reco-η fiducial loss** ($\sim$1.6%) is intrinsic to the choice of $|\eta_{\text{reco}}| < 0.7$. If the analysis can tolerate $|\eta_{\text{reco}}| < 0.75$ and migrate the truth fiducial in lockstep, this contribution shrinks. Not free — the cross-section would change to reflect the wider acceptance.
4. **No-`trkID`-match** ($\sim$1.7%) is the hardest to attack without a re-tuned cluster-truth association. The macro's choice of `trkID`-only matching (no ΔR fallback) is documented as deliberate.
5. **Add the signal-side breakdown histograms** so future $\varepsilon_{\text{reco}}$ studies can be done directly from `MC_efficiency_*.root` without ad-hoc Python scripts.

Files referenced

- Plot: `plotting/figures/photon_reco_eff_bdt_nom.pdf`

- Macro: `plotting/plot_reco_eff.C`
- Breakdown JSON: `reports/reco_eff_breakdown.json`
- Breakdown script: `reports/reco_eff_breakdown.py`
- Source: `efficiencytool/results/MC_efficiency_bdt_nom.root::eff_reco_eta_0`
- Cuts/config: `efficiencytool/config_bdt_nom.yaml` (`reco_min_ET=5.0`, etc.)
- Macro definition: `efficiencytool/RecoEffCalculator_TTreeReader.C` (truth selection 1672–1739, cluster loop 2062–2926, eff fill 2933–2978)